Nonradial mass-critical biharmonic NLS

Verified obstructions and partial finite-time blowup theorems

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Abstract

We investigate whether every negative-energy $H^2(\mathbb{R}^N)$ solution of the focusing mass-critical biharmonic nonlinear Schrödinger equation blows up in finite time, without radial symmetry. The estimates below concern the preliminary virial approach; the later excited-wave construction gives global negative-energy nonradial solutions. We prove that the proposed unrestricted small-coefficient weighted interpolation estimate is false, even at fixed mass, energy -1, and zero momentum, and even with an optimally chosen localization center. We replace it by a weighted estimate involving mass in relative balls. Combined with the exponential virial identities, this gives a conditional blowup criterion and unconditional consequences for several invariant, genuinely nonradial symmetry classes. A putative global negative-energy solution must develop a precise form of angular concentration. A separate angular-regularity criterion and traveling-wave tests identify further constraints. The general virial mechanism is inherited from prior work; the additional estimates concern the explicitly stated symmetry and concentration classes.

1 Problem, status, and provenance

Set $N \geq 2$ and $p = 2 + 8/N$. The equation and conserved quantities are

$$i\partial_t u = \Delta^2 u - |u|^{p-2}u, \qquad u(0) = u_0 \in H^2(\mathbb{R}^N), \tag{1}$$

$$m = M(u) = \|u\|_2^2, \qquad E(u) = \frac{1}{2}\|\Delta u\|_2^2 - \frac{1}{p}\|u\|_p^p, \qquad P(u) = \operatorname{Im}\int \bar{u}\nabla u\, dx. \tag{2}$$

Let $(-T_-, T_+)$ be the maximal H^2 lifespan. The target is

$$E(u_0) < 0 \quad \Longrightarrow \quad T_+ < \infty \text{ and } T_- < \infty \tag{3}$$

for arbitrary initial data, with no symmetry or finite-variance hypothesis. Finite lifespan means divergence of the H^2 norm through the usual blowup alternative. Infinite-time growth of that norm is not sufficient to prove (3).

Boulenger–Lenzmann [1] obtained the radial finite-time/infinite-time alternative. The manuscript posted on 2 October 2026 by Dinh [2] proves finite-time blowup for radial data in every $N \geq 2$. Its Remark 1.4 explicitly retains the nonradial question. The present literature check did not locate a full resolution under (1)–(3). Results with an inhomogeneous coefficient, extra dispersion, or additional symmetry do not directly answer this formulation.

The statements below distinguish three levels: established source identities; derivations proved in this note; and unresolved requirements. The proofs of the local-mass estimate, orbit-packing consequence, and static obstructions were cross-checked by independent derivations during this research session. This is not a substitute for external mathematical review or a proof of publication novelty.

2 The part of the virial argument that needs no symmetry

For a real smooth multiplier ϕ with the required bounded derivatives, define

$$V_\phi(u) = 2\,\mathrm{Im}\int \bar{u}\nabla\phi\cdot\nabla u\,dx.$$

Direct differentiation, integration by parts, and (1) give

$$\begin{aligned} V_\phi' = {} & 8\,\mathrm{Re}\sum_{i,j,k}\int \phi_{jk}\bar{u}_{ij}u_{ik}\,dx - 4\,\mathrm{Re}\sum_{j,k}\int(\Delta\phi)_{jk}\bar{u}_j u_k\,dx \\ & - 2\int \Delta^2\phi|\nabla u|^2\,dx + \int \Delta^3\phi|u|^2\,dx - \frac{8}{N+4}\int \Delta\phi|u|^p\,dx. \end{aligned} \tag{4}$$

For H^2 solutions this is understood via smooth approximation, or as the integrated localized multiplier identity. Each displayed right-hand term is continuous in H^2 for fixed multiplier. The local theory supplies the corresponding approximation on compact lifespan intervals.
Fix a center c and $R>0$. Write $r=|x-c|$, $s=r/R$, and choose

$$h = e^{-s^4}, \quad w = 1-h, \quad \nabla\phi_{R,c} = (x-c)h, \quad D = \int w|D^2u|_F^2\,dx, \quad L = \int w|u|^p\,dx. \tag{5}$$

The quartic exponent in this weight is important. It is not the Gaussian weight e^{-s^2}. The multiplier Hessian is

$$D^2\phi_{R,c} = hI - 4s^4h\,e_r\otimes e_r.$$

Thus the exact principal and nonlinear decomposition is

$$V_R' = 16E - 8D - 32\mathcal{H} + B + \mathcal{L}, \tag{6}$$

where

$$\begin{aligned} \mathcal{H} &= \int s^4h|D^2u\,e_r|^2\,dx \ge 0, \\ B &= \frac{8}{N+4}\int [Nw+4s^4h]|u|^p\,dx, \qquad 0\le B\le 8L, \\ \mathcal{L} &= -4\,\mathrm{Re}\sum_{j,k}\int(\Delta\phi_{R,c})_{jk}\bar{u}_j u_k\,dx - 2\int\Delta^2\phi_{R,c}|\nabla u|^2\,dx + \int\Delta^3\phi_{R,c}|u|^2\,dx. \end{aligned}$$

We used $s^4e^{-s^4}\le 1-e^{-s^4}$ and the unweighted identity $\|D^2u\|_{L^2,F}^2 = \|\Delta u\|_2^2$.
Let $g=\sqrt{w}$. Near the origin g is a smooth multiple of r^2/R^2; hence

$$\begin{aligned} \|\Delta g\|_\infty &\le C_N R^{-2}, \\ |D^2\Delta\phi_{R,c}| + |\Delta^2\phi_{R,c}| &\le C_N R^{-2}g, \qquad |\Delta^3\phi_{R,c}| \le C_N R^{-4}. \end{aligned}$$

Integration by parts gives

$$\begin{aligned} \int g|\nabla u|^2\,dx &= -\,\mathrm{Re}\int g\bar{u}\Delta u\,dx + \frac{1}{2}\int\Delta g|u|^2\,dx \\ &\le \sqrt{N}\,m^{1/2}D^{1/2} + C_N R^{-2}m. \end{aligned} \tag{7}$$

Consequently, for every $\eta>0$,

$$|\mathcal{L}| \le \eta D + C_{N,\eta}R^{-4}m. \tag{8}$$

Finally $|\nabla\phi_{R,c}|^4 \le C_N R^4 w$, Cauchy–Schwarz, and (7) imply

$$|V_R(u)| \le C_N m + C_N R m^{3/4}D^{1/4}. \tag{9}$$

Equations (6)–(9) require no radial symmetry. They are the inherited exponential virial mechanism. The missing input is control of L.

3 Why direct unrestricted absorption cannot work

3.1 A translated concentration

For $a \neq c$, $f \in C_c^\infty(\mathbb{R}^N) \setminus \{0\}$, put

$$f_{\lambda,a}(x) = \lambda^{N/2} f(\lambda(x-a)).$$

Mass is independent of λ, while, with R, c, a fixed,

$$\lambda^{-4} \int w |f_{\lambda,a}|^p \, dx \longrightarrow w(a) \, \|f\|_p^p ,$$
$$\lambda^{-4} \int w |D^2 f_{\lambda,a}|_F^2 \, dx \longrightarrow w(a) \left\| D^2 f \right\|_2^2 .$$

Therefore an estimate $L \leq \varepsilon D + C_{\varepsilon,m} R^{-4}$ for arbitrary $\varepsilon > 0$ and all H^2 data would force $\|f\|_p^p \leq \varepsilon \|D^2 f\|_2^2$ for arbitrary ε, a contradiction.

3.2 Fixed invariants and an optimally chosen center

Proposition 3.1. *For every fixed $R > 0$ there are $m_0 > 0$ and real $U_\lambda \in C_c^\infty(\mathbb{R}^N)$ such that*

$$M(U_\lambda) = m_0, \qquad E(U_\lambda) = -1, \qquad P(U_\lambda) = 0,$$

and, uniformly over all centers $c \in \mathbb{R}^N$,

$$\frac{L_{R,c}(U_\lambda)}{D_{R,c}(U_\lambda)} \longrightarrow \frac{p}{2}, \qquad \inf_c D_{R,c}(U_\lambda) \longrightarrow \infty.$$

In particular, choosing a center separately for each datum cannot give $L_{R,c} \leq \varepsilon D_{R,c} + C$ for any $\varepsilon < p/2$, even if C depends on the fixed mass, energy, momentum, and R.

Proof. Choose real radial $q \in C_c^\infty \setminus \{0\}$ and $A > 0$ so that $f = Aq$ has zero energy. Explicitly,

$$A^{p-2} = \frac{p \, \|\Delta q\|_2^2}{2 \, \|q\|_p^p}.$$

Then $K := \|D^2 f\|_2^2 > 0$ and $Q := \|f\|_p^p = (p/2)K$. Choose a real compactly supported b with $E(b) = -1$: first increase amplitude to make energy negative, then use critical spatial scaling to normalize it. Choose distinct points a_1, a_2 outside the support of b and put

$$U_\lambda = f_{\lambda,a_1} + f_{\lambda,a_2} + b.$$

For all large λ the three supports are disjoint. Thus the invariant identities hold exactly, with $m_0 = 2M(f) + M(b)$.
Set $W_c = w_{R,c}(a_1) + w_{R,c}(a_2)$ and $d = |a_1 - a_2|$. At least one point is at distance $d/2$ or more from c, so

$$W_c \geq 1 - e^{-(d/(2R))^4} > 0.$$

The translated weights satisfy $\|\nabla w_{R,c}\|_\infty \leq C/R$ uniformly in c. Changing variables and using the mean value theorem yields

$$L_{R,c}(U_\lambda) = \lambda^4 W_c Q + O_R(\lambda^3) + O(1), \quad D_{R,c}(U_\lambda) = \lambda^4 W_c K + O_R(\lambda^3) + O(1),$$

with both remainders uniform in c. Division proves the claim. □

This proposition is an obstruction to a static estimate, not a global solution and not a counterexample to (3). It does not rule out a scale R that changes with the datum or time, provided the resulting new terms can be controlled.

3.3 Keeping the signed terms does not recover the same defect

Let $\mathcal{V}_\phi(u)$ denote the right-hand side of (4). For the real radial zero-energy f above,

$$\mathrm{Re}\sum_i \int \bar{f}_{ij} f_{ik}\, dx = \frac{K}{N}\delta_{jk}.$$

The leading λ^4 contribution of one packet to $\mathcal{V}_\phi$ is

$$\lambda^4 \Delta\phi(a)\left(\frac{8K}{N} - \frac{8Q}{N+4}\right) = 0.$$

The error is $O_R(\lambda^3)$ for translated exponential multipliers; lower-order terms are $O_R(\lambda^2)$. Applying this to both packets in Proposition 3.1 shows that a universal estimate

$$\mathcal{V}_{\phi_{R,c}}(u) \le 16E(u) - \gamma D_{R,c}(u) + C, \qquad \gamma > 0,$$

also fails on this fixed-invariant class, even with optimized c. The defect grows as λ^4, while $\mathcal{V}$ has no contribution of that order. Rearranging the same signed terms is therefore insufficient.

4 A valid local-mass replacement

For $0 < \delta < 1/4$ and fixed c, define

$$\mu_{\delta,c}(u) = \sup_{a \ne c} \int_{B(a,\delta|a-c|)} |u(x)|^2\, dx. \tag{10}$$

These balls are narrow relative to their distance from the center. This quantity measures concentration away from a single distinguished point.

Lemma 4.1. *There is C_N independent of u, R, c, δ such that*

$$L_{R,c}(u) \le C_N \mu_{\delta,c}(u)^{4/N}\left[D_{R,c}(u) + \delta^{-4}R^{-4}M(u)\right], \qquad u \in H^2(\mathbb{R}^N). \tag{11}$$

Proof. First work on $B(a,r)$. Put $m_a = \int_{B(a,4r)} |u|^2$ and $H_a = \int_{B(a,4r)} |D^2u|_F^2$. Choose smooth cutoffs $\eta = 1$ on $B(a,r)$ supported in $B(a,2r)$ and $\chi = 1$ on $B(a,2r)$ supported in $B(a,4r)$. Their derivatives have the natural powers of r^{-1}. Writing $J = \int \chi^2|\nabla u|^2$, integration by parts gives

$$J \le \sqrt{N}\, m_a^{1/2} H_a^{1/2} + C_N r^{-1} m_a^{1/2} J^{1/2}.$$

Absorption and Young's inequality yield

$$r^{-2}\int_{B(a,2r)} |\nabla u|^2 \le C_N H_a + C_N r^{-4} m_a.$$

The product rule and the mass-critical Gagliardo–Nirenberg inequality imply

$$\int_{B(a,r)} |u|^p \le \|\eta u\|_p^p \le C_N m_a^{4/N}(H_a + r^{-4}m_a). \tag{12}$$

Translate so $c = 0$. On each annulus $A_j = \{2^j \le |x| < 2^{j+1}\}$, choose a maximal r_j-separated family of centers with $r_j = \delta 2^j/64$. Balls of radius r_j cover A_j. Balls with radius $4r_j$ have bounded overlap, with a bound depending only on N, both within each annulus and across all annuli. Each enlarged ball is contained in $B(a, \delta|a|)$, so $m_a \le \mu_{\delta,0}(u)$.

On every enlarged ball, $w_R(x)$ is comparable to $w_R(a)$ with constants independent of δ, R, j. This follows from $w_R(x) \asymp \min\{|x|^4/R^4, 1\}$. Also

$$w_R(a) r_j^{-4} \le C_N \delta^{-4} R^{-4}.$$

Multiply (12) by $w_R(a)$ and sum over the cover. Bounded overlap controls the Hessian sum by $D_{R,0}$ and the cutoff sum by $C_N\delta^{-4}R^{-4}m$. This proves (11). The cutoff argument is valid directly in H^2, or by smooth approximation. □

5 Conditional finite-time blowup and a necessary concentration pattern

Theorem 5.1. *There exists $m_*(N) > 0$ such that a maximal solution with $E(u_0) < 0$ has $T_+ < \infty$ if some fixed c and some fixed $\delta \in (0, 1/4)$ satisfy*

$$\sup_{0 \le t < T_+} \mu_{\delta,c}(u(t)) \le m_*(N). \tag{13}$$

An analogous backward-time statement holds.

Proof. Take $\eta = 2$ in (8), and choose m_* so that the coefficient $C_N m_*^{4/N}$ in (11) is at most $1/4$. Then

$$V_R' \le 16E - 4D + C_N R^{-4} m (1 + \delta^{-4} m_*^{4/N}).$$

Choose one sufficiently large, fixed R to obtain

$$V_R' \le -8|E| - 4D. \tag{14}$$

Suppose the solution were forward global. With $y = -V_R$, (14) makes y reach $2C_N m$ after finite time. Thereafter (9) implies

$$D \ge c_N R^{-4} m^{-3} y^4, \qquad y' \ge c_N R^{-4} m^{-3} y^4.$$

Integrating the differential inequality for y^{-3} gives a contradiction in finite time. Backward time uses $\widetilde{u}(t,x) = \overline{u(-t,x)}$, which satisfies the same equation with the same mass, energy, and relative-ball masses. □

The hypothesis is dynamical. It cannot be inferred from negative energy alone by the estimates established here.

Corollary 5.2. *If a negative-energy solution is forward global, then for every fixed c there exist $t_n \to \infty$, $\delta_n \downarrow 0$, and $a_n \ne c$ such that*

$$\int_{B(a_n, \delta_n |a_n - c|)} |u(t_n, x)|^2 \, dx \ge \frac{1}{2} m_*(N).$$

Proof. For every fixed $f \in L^2$, $\mu_{\delta,c}(f) \to 0$ as $\delta \downarrow 0$. Indeed, centers very far from c give balls in a small spatial tail, while centers in a bounded region give sets of uniformly vanishing volume. Absolute continuity of the mass integral handles the latter. Moreover

$$|\mu_{\delta,c}(f) - \mu_{\delta,c}(g)| \le (\|f\|_2 + \|g\|_2) \, \|f - g\|_2 .$$

The convergence is therefore uniform on compact subsets of L^2, hence on each compact time interval of the solution. The contrapositive of Theorem 5.1 supplies, for every δ, a time and a relative ball carrying more than $m_*/2$. Choose δ_n recursively smaller than 2^{-n} and $\delta_{n-1}/2$, and so small that this cannot happen before time n; then choose such a ball. No attainment of the supremum is needed. □

This identifies a scenario that a general proof must exclude or control. It does not itself exclude that scenario. Negative energy supplies concentration somewhere; it does not supply the uniform upper bound in (13).

6 Unconditional results in invariant nonradial classes

Theorem 6.1 (Infinite-orbit symmetry)**.** *Let $G \subset O(N)$ be compact, with Gx infinite for every $x \neq 0$. Let $\chi : G \to S^1$ be a continuous unitary character. Suppose*

$$u_0(gx) = \chi(g)u_0(x), \qquad g \in G, \qquad u_0 \in H^2(\mathbb{R}^N), \quad E(u_0) < 0.$$

Then the maximal solution has $T_+, T_- < \infty$.

Proof. Rotation invariance, phase invariance, and uniqueness preserve the equivariance at every time. In particular the solution's mass density is G-invariant.

Fix any integer K. Every unit-vector orbit contains K distinct points. For each unit vector choose K group elements realizing positive minimum pairwise separation. The same separation, with a factor $1/2$, holds in a neighborhood of that vector. A finite cover of the unit sphere gives a uniform $d_K > 0$ such that every orbit has K points separated by at least $d_K|x|$.

For $\delta < \min\{1/4, d_K/2\}$ the corresponding K balls of radius $\delta|x|$ are disjoint and have equal mass. Hence $\mu_{\delta,0}(u(t)) \leq m/K$ at every time. Choose K so large that $m/K \leq m_*(N)$ and apply Theorem 5.1 in both time directions. □

Only initial *density* symmetry is not enough: it need not be preserved. The theorem imposes symmetry on the complex field, with a compatible phase character.

Planar vortices. In $N = 2$, take $G = SO(2)$ and $\chi(R_\theta) = e^{i\ell\theta}$, $\ell \in \mathbb{Z}$. This gives the rigorous implication

$$u_0(r, \theta) = e^{i\ell\theta}v_0(r) \in H^2(\mathbb{R}^2), \quad E(u_0) < 0 \quad \Longrightarrow \quad T_+, T_- < \infty.$$

For $\ell \neq 0$ the complex field is nonradial. The class is nonempty: multiply $(x_1 + ix_2)^\ell e^{-|x|^2}$ for $\ell \geq 0$, or $(x_1 - ix_2)^{|\ell|}e^{-|x|^2}$ for $\ell < 0$, by a sufficiently large positive amplitude. This remains a restricted class and does not address arbitrary planar data.

Biradial data in all $N \geq 4$. Take $G = O(2) \times O(N - 2)$ and write $x = (y, z)$. Then $u_0(x) = v_0(|y|, |z|)$ is allowed. Every nonzero orbit is infinite because at least one block has nonzero length and both block dimensions are at least two. A genuinely nonradial example is $u_0 = Ae^{-|y|^2-2|z|^2}$ for sufficiently large A. No finite-variance hypothesis is needed in the theorem, even though this example has finite variance.

Finite cyclic symmetry. For $N = 2d$ and an integer $K \geq 2$, identify $\mathbb{R}^N = \mathbb{C}^d$ and put $g(z) = e^{2\pi i/K}z$. If $u_0(gx) = \zeta u_0(x)$ with $\zeta^K = 1$, then for $\delta < \min\{1/4, \sin(\pi/K)\}$ the K orbit balls are disjoint. Thus

$$E(u_0) < 0, \qquad M(u_0)/K \leq m_*(N) \quad \Longrightarrow \quad T_+, T_- < \infty.$$

Here only a finite symmetry is imposed, but a mass-to-orbit-size condition is required. These statements are consequences proved in this note; their external novelty has not been established.

7 A complementary angular-regularity criterion

Define, when finite,

$$K(u) = \int_0^\infty \|(1 - \Delta_{S^{N-1}})u(r, \cdot)\|^2_{L^2(S^{N-1})}\, r^{N-1}\, dr.$$

This angular H^2 mass satisfies $K \geq m$. It is not automatically finite for arbitrary $H^2(\mathbb{R}^N)$ data and is not generally conserved.

Proposition 7.1. *For every $\varepsilon > 0$,*

$$L_{R,0}(u) \le \varepsilon D_{R,0}(u) + C_{N,\varepsilon} R^{-4} \left[K(u) m^{4/(N-1)} + K(u)^{(N-1)/N} m^{5/N} \right]. \tag{15}$$

Consequently a global negative-energy solution cannot satisfy

$$\limsup_{T\to\infty} \frac{1}{T} \int_0^T K(u(t))\, dt < \infty.$$

Proof. On a fixed annular cylinder, let v be a localized rescaled function and set $\mu = \|v\|_2^2$, $Q = \|\partial_r^2 v\|_2^2 + \mu$, and $A = \|v\|_{L_r^2 H_\theta^2}^2$, using equivalent norms with $A \ge \mu$. The anisotropic inequality is

$$\|v\|_p^p \le C_N Q^{1/N} A^{(N-1)/N} \mu^{4/N}. \tag{16}$$

To verify it, use finitely many smooth sphere charts and Fourier variables $(\xi, \eta) \in \mathbb{R} \times \mathbb{R}^{N-1}$. Hausdorff–Young and weighted Hölder with $W = 1 + (\xi/a)^4 + |\eta/b|^4$ give

$$\|v\|_p \le C \left(\int W |\widehat{v}|^2 \right)^{1/2} \left(\int W^{-p/(p-2)} \right)^{1/2-1/p}.$$

The integral is finite since $p/(p-2) = (N+4)/4 > N/4$. Put $a = (Q/\mu)^{1/4}$ and $b = (A/\mu)^{1/4}$; the case $\mu = 0$ is immediate. Raising to the power p gives (16). Chart and interval cutoff derivatives are controlled by their fixed-domain H^2 norms.

On dyadic annuli of radius $\rho_j = 2^j R$, let $\alpha_j = \rho_j^{-4} \min\{1, (\rho_j/R)^4\} \le R^{-4}$. Write D_j, A_j, μ_j for the weighted Hessian, angular mass, and mass on fixed enlarged annuli. Rescaling and the cutoff gradient estimate used in Lemma 4.1 give

$$\begin{aligned} \int_{A_j} w|u|^p \le{}& C\alpha_j^{1-1/N} D_j^{1/N} A_j^{(N-1)/N} \mu_j^{4/N} \\ &+ C\alpha_j A_j^{(N-1)/N} \mu_j^{5/N}. \end{aligned}$$

Young's inequality bounds the first term by $\varepsilon' D_j + C_{\varepsilon'} \alpha_j A_j \mu_j^{4/(N-1)}$. Sum with bounded overlap and use

$$\sum_j A_j^{(N-1)/N} \mu_j^{5/N} \le \left(\sum_j A_j \right)^{(N-1)/N} \left(\sum_j \mu_j^5 \right)^{1/N}.$$

This proves (15).

For the time-average assertion, $K \ge m$ implies $K^{(N-1)/N} m^{5/N} \le K m^{4/N}$. Combine (15) with the virial bounds, integrate in time, and choose one fixed large R using the supposed finite upper average of K. For $y = -V_R$ this gives

$$y(t) \ge 8|E|t - C + c \int_0^t D(s)\, ds.$$

The bridge (9) implies $D \ge c_R (y - C_N m)_+^4$. The right-hand integral lower bound therefore satisfies a superlinear integral differential inequality once positive. More explicitly, define $F(t) = 8|E|t - C + c\int_0^t D$; then $y \ge F$, and for all sufficiently large t, $F' \ge c_R (F - C_N m)^4$. This contradicts global existence. □

For a planar ℓ-vortex, $K = (1+\ell^2)^2 m$ is constant; this independently recovers that case of Theorem 6.1. Conserved total angular momentum, a much weaker signed quantity, cannot replace K.

Corollary 7.2 (Quantitative angular growth)**.** *If a forward-global negative-energy solution has locally integrable $K(t) = K(u(t))$, then there exist $c(u_0, N) > 0$ and T_0 such that*

$$\int_0^T K(t)\,dt \geq c(u_0, N)T^5, \qquad T \geq T_0.$$

Proof. Put $e = |E|$ and $S(T) = \int_0^T K$. The preceding estimates, using $K \geq m > 0$, give

$$y(t) \geq -V_R(u_0) + 16et + c_N \int_0^t D(s)\,ds - C_{N,m}R^{-4}S(t).$$

Also $|V_R(u_0)| \leq C(u_0)R$ by Cauchy–Schwarz and boundedness of $\nabla\phi_R/R$. For each large test horizon T, choose $R = \alpha T$, which remains fixed throughout $[0, T]$. First choose $\alpha > 0$ small enough that $C(u_0)\alpha \leq e$. If $S(T) \leq \kappa T^5$, choose $\kappa > 0$ small enough that $C_{N,m}\alpha^{-4}\kappa \leq e$. Then, for $T/2 \leq t \leq T$,

$$y(t) \geq 6eT + c_N \int_{T/2}^t D(s)\,ds =: F(t).$$

For sufficiently large T, $F \geq 2C_N m$. The bridge gives $F' \geq c_{N,m}\alpha^{-4}T^{-4}F^4$. Starting at $F(T/2) = 6eT$, this forces divergence within $C_{N,m,e}\alpha^4 T$. Choosing α smaller, before choosing κ, makes this time less than $T/2$, a contradiction. Thus $S(T) > \kappa T^5$ for every sufficiently large T. No derivative of a time-dependent R was used. □

8 Additional proof obstructions and counterexample tests

8.1 A direct fixed-multiplier Riccati inequality can have the wrong sign

It is also possible to have very negative V_R but very positive instantaneous V_R' at fixed mass, energy -1, and momentum zero. Here is a reproducible algebraic construction.
Choose real radial $q \in C_c^\infty \setminus \{0\}$, $k \neq 0$, and $A > 0$ so $f = Ae^{ik\cdot x}q$ has zero energy. Put $m_q = \|q\|_2^2$, $a_q = \|\nabla q\|_2^2/N$, and $b_k = (N+4)a_q + |k|^2 m_q$. The traceless integrated kinetic tensor is

$$\mathrm{Re}\sum_i \int \bar{f}_{ij} f_{ik} - \frac{\|\Delta f\|_2^2}{N}\delta_{jk} = A^2 b_k \left(k_j k_k - \frac{|k|^2}{N}\delta_{jk}\right).$$

This identity follows by shifting the radial Fourier density by k and expanding its fourth moments. Place critically scaled copies at $\pm ae$, with modulations k and $-k$, where

$$a > 0, \quad |e| = 1, \qquad -|k|/\sqrt{N} < k \cdot e < 0.$$

More explicitly, add a fixed disjoint real packet b of energy -1 and set

$$U_\lambda(x) = A\lambda^{N/2}[e^{i\lambda k\cdot(x-ae)}q(\lambda(x-ae)) + e^{-i\lambda k\cdot(x+ae)}q(\lambda(x+ae))] + b(x).$$

The physical modulations are $\pm\lambda k$. For the multiplier centered at zero,

$$V_R(U_\lambda) = 4\lambda a e^{-a^4/R^4} A^2 m_q (k\cdot e) + O(1) \to -\infty.$$

Write $h_r = e^{-s}(1-4s)$ and $h_t = e^{-s}$ with $s = a^4/R^4$. Each concentrated packet contributes

$$8\lambda^4(h_r - h_t)A^2 b_k \left((k\cdot e)^2 - |k|^2/N\right) + O_R(\lambda^3) > 0$$

to the derivative for large λ. Thus $\mathcal{V}_{\phi_R}(U_\lambda) \to +\infty$. A universal instantaneous negative quartic inequality for this fixed V_R cannot hold without further information.

8.2 Moving centers and traveling profiles

If $c = c(t)$, differentiation adds the term

$$-2\dot{c}(t) \cdot \operatorname{Im} \int \bar{u}\, D^2 \phi_{R,c(t)} \nabla u \, dx.$$

It cannot simply be discarded. Total momentum zero does not force each separated component to have zero momentum. Fourth-order dispersion also lacks the usual second-order Galilean symmetry.

A possible genuine counterexample would be a traveling wave $u(t,x) = e^{i\omega t} Q(x - vt)$ with $Q \in H^2$ and $E(Q) < 0$. Its profile solves

$$\Delta^2 Q + \omega Q + iv \cdot \nabla Q - |Q|^{p-2} Q = 0.$$

For a sufficiently localized profile, differentiating $E + \omega M/2 - v \cdot P/2$ along mass-preserving scaling gives

$$4E(Q) - \tfrac{1}{2} v \cdot P(Q) = 0, \qquad E(Q) = \tfrac{1}{8} v \cdot P(Q).$$

Standing profiles have zero energy. This identity does not by itself determine the sign for arbitrary excited traveling profiles.

Exploratory one-dimensional and planar profile computations were undertaken in the preliminary calculations. A 14-pulse one-dimensional candidate at velocity $v = 0.01$ gave energy approximately -4.35117×10^{-7}, with $E - vP/8$ approximately -1.84×10^{-13} in a finite-domain boundary-value calculation. This is a candidate, not a certified real-line solution. Planar collinear and square multipulse tests did not produce a negative response form. These calculations do not answer the specified $N \geq 2$ problem. Domains, boundary conditions, solver limitations, data, and code are retained in the accompanying research bundle; the proved statements above do not depend on these computations.

9 Relation to the subsequent construction

The static obstructions explain why conserved mass, energy and momentum do not close the proposed multiplier argument. The later manuscript *Global negative-energy traveling waves for the mass-critical biharmonic Schrödinger equation in four space dimensions* constructs an exact excited traveling wave, and the three-dimensional extension gives a second example. Thus the unconditional nonradial negative-energy collapse assertion is false. The conditional virial estimates and symmetry-restricted theorems above remain valid. The remaining dynamical question concerns general near-critical trajectories, rather than an unconditional blowup statement.

References

[1] T. Boulenger and E. Lenzmann, *Blowup for Biharmonic NLS*, arXiv:1503.01741, version 2, 21 April 2015. https://arxiv.org/abs/1503.01741.

[2] L. Dinh, *Finite-Time Blow-Up of Radial Negative-Energy Solutions for the Mass-Critical Biharmonic Nonlinear Schrödinger Equation*, 20-page manuscript, Meta AI Research, posted 2 October 2026. Official publication page and full PDF.